25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Vectors & Embeddings in AI Practice Exercises

1 MCQ

Compute the Manhattan (L_1) norm and Euclidean (L_2) norm of the word vector $\mathbf{w} = [10, -15]^T$.

A $\|\mathbf{w}\|_1 = \sqrt{325},\ \|\mathbf{w}\|_2 = 25$

B $\|\mathbf{w}\|_1 = -5,\ \|\mathbf{w}\|_2 = 325$

C $\|\mathbf{w}\|_1 = 25,\ \|\mathbf{w}\|_2 = 25$

D $\|\mathbf{w}\|_1 = 25,\ \|\mathbf{w}\|_2 = \sqrt{325}$

2 MCQ

Two document embeddings $\mathbf{a} = [1, 2]^T$ and $\mathbf{b} = [2, 1]^T$ are compared. If $\|\mathbf{a}\| = \|\mathbf{b}\| = \sqrt{5}$, calculate their exact cosine similarity.

A 1.000

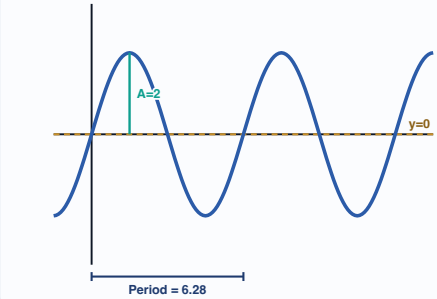
B 0.640

C 0.800

D 0.500

3 MCQ

Two document embeddings $\mathbf{a} = [2, 4]^T$ and $\mathbf{b} = [4, 2]^T$ are compared. If $\|\mathbf{a}\| = \|\mathbf{b}\| = \sqrt{20}$, calculate their exact cosine similarity.



A 1.000

B 0.640

C 0.800

D 0.500

4 MCQ

Two document embeddings $\mathbf{a} = [3, 6]^T$ and $\mathbf{b} = [6, 3]^T$ are compared. If $\|\mathbf{a}\| = \|\mathbf{b}\| = \sqrt{45}$, calculate their exact cosine similarity.

A 1.000

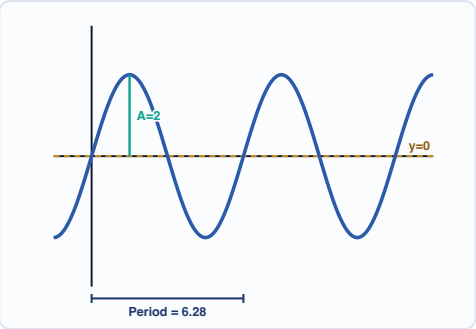
B 0.640

C 0.800

D 0.500

5 MCQ

Two document embeddings $\mathbf{a} = [4, 8]^T$ and $\mathbf{b} = [8, 4]^T$ are compared. If $\|\mathbf{a}\| = \|\mathbf{b}\| = \sqrt{80}$, calculate their exact cosine similarity.



A 1.000

B 0.640

C 0.800

D 0.500

6 MCQ

Two document embeddings $\mathbf{a} = [5, 10]^T$ and $\mathbf{b} = [10, 5]^T$ are compared. If $\|\mathbf{a}\| = \|\mathbf{b}\| = \sqrt{125}$, calculate their exact cosine similarity.

A 1.000

B 0.640

C 0.800

D 0.500

7 MCQ

Two document embeddings $\mathbf{a} = [6, 12]^T$ and $\mathbf{b} = [12, 6]^T$ are compared. If $\|\mathbf{a}\| = \|\mathbf{b}\| = \sqrt{180}$, calculate their exact cosine similarity.

A 1.000

B 0.640

C 0.800

D 0.500

8 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 6x + 20$ in the domain of Vectors & Embeddings in AI. By completing the square $f_1(x) = (x - h)^2 + k$, find the sum $k + f_1(5)$.

A 26

B 29

C 15

D 24

9 MCQ

Consider two vectors, $\mathbf{u} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. What is the vector representing $2\mathbf{u} - \mathbf{v}$?

A $\begin{bmatrix} 2 \\ 12 \end{bmatrix}$

B $\begin{bmatrix} 5 \\ 6 \end{bmatrix}$

C $\begin{bmatrix} 5 \\ 10 \end{bmatrix}$

D $\begin{bmatrix} 7 \\ 10 \end{bmatrix}$

10 MCQ

A vector $\mathbf{v}$ is transformed into a new vector $\mathbf{v}'$ by a linear transformation represented by the matrix $\mathbf{M}$. We are given that $\mathbf{v} = (2\ 3)$ and $\mathbf{v}' = (7\ 11)$. The transformation matrix is of the form $\mathbf{M} = (a \quad 1\ 4 \quad b)$. Determine the value of a and b.

A a = 3, b = 2

B a = 2, b = 1

C a = 4, b = -1

D a = 1, b = 3

11 MCQ

Examining a dataset, the prior probability of class C_1 is $P(C_1) = 0.60$. What is the probability of the complement class $P(\neg C_1)$?

A 0.70

B 0.40

C 0.30

D 1.60

12 MCQ

Principal Component Analysis on covariance matrix Σ_X yields eigenvalues $\lambda_1 = 83$ and $\lambda_2 = 17$. Determine the percentage of variance h_u explained by the first principal component.

A 87

B 63

C 88

D 83

13 MCQ

Consider a dataset of n data points, where each data point is a vector in R^D. A non-linear dimensionality reduction technique maps these data points to a lower-dimensional space R^d. The mapping function for a single data point $\mathbf{x} \in \mathrm{R}^D$ is given by $f(\mathbf{x}) = \phi(\mathbf{x})$, where $\phi : \mathrm{R}^D \to \mathrm{R}^d$ is a function designed to preserve local neighborhood structures. Suppose the mapping ϕ is defined by an explicit feature map $\phi(\mathbf{x}) = [\mathbf{x}^T\mathbf{x}, \mathbf{x}^T\mathbf{x}, \ldots, \mathbf{x}^T\mathbf{x}]^T$ where the vector $\mathbf{x} \in \mathrm{R}^{10}$ and d is the dimension of the embedded space. If the goal is to embed these points into a space such that pairwise Euclidean distances are preserved as closely as possible under a specific distortion, and it is found that $d = 15$ yields optimal results for this mapping under a certain constraint, determine the effective rank of the feature map's image space.

A 15

B 1

C 10

D 25

14 MCQ

A binary classifier $h(u)$ produces the confusion matrix below with $\mathrm{TP} = 60$, $\mathrm{FP} = 12$, $\mathrm{FN} = 6$, and $\mathrm{TN} = 10$. Determine the total number of correct classifications $k = \mathrm{TP} + \mathrm{TN}$.

A 70

B 64

C 74

D 85

15 MCQ

Consider the scalar loss function $L(w) = 3w^2 - 8w + 5$. Find the analytical gradient $\nabla L(w)$ evaluated at $w = 4$.

A $\nabla L(4) = 24$

B $\nabla L(4) = 16$

C $\nabla L(4) = 32$

D $\nabla L(4) = 4$

16 MCQ

Suppose a machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.20$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

A 1.1000

B 0.7000

C 0.0200

D 0.1800

17 MCQ

Suppose a 3D dataset has covariance matrix eigenvalues $\lambda_1 = 68$, $\lambda_2 = 21$, and $\lambda_3 = 11$. What percentage of total variance is explained by the first principal component PC_1?

A 89.0%

B 68.0%

C 21.0%

D 76.4%

18 MCQ

Consider the sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 1.20$, what is the value of the derivative $\sigma'(z_0)$?

A -0.2000

B 1.4400

C -0.2400

D 2.4000

19 MCQ

Considering anomaly detection, anomaly prior $P(A) = 0.04$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|\neg A) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly?

A 0.221

B 0.442

C 0.950

D 0.619

20 MCQ

For a bivariate feature dataset, $\mathrm{Cov}(x, y) = 72$, $\mathrm{Var}(x) = 24$, $\bar{x} = 5$, and $\bar{y} = 50$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

A $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 50$

B $\hat{\beta}_1 = 4,\ \hat{\beta}_0 = 30$

C $\hat{\beta}_1 = \frac{1}{3},\ \hat{\beta}_0 = 35$

D $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 35$

21 MCQ

A centered feature matrix $\mathbf{X}$ has shape $N \times 18$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$?

A 18×18 symmetric matrix

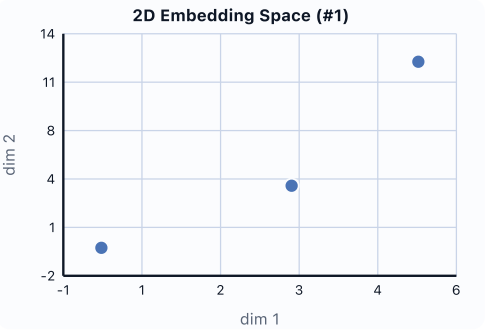
B $N \times N$ symmetric matrix

C 18×18 skew-symmetric matrix

D $N \times 18$ rectangular matrix

22 MCQ

Compute the cosine similarity $\cos(\theta) = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|_2\|\mathbf{v}\|_2}$ between embedding vectors $\mathbf{u} = [3, 4]^\top$ and $\mathbf{v} = [5, 12]^\top$ (embedding pair #1).



A $\frac{66}{65}$

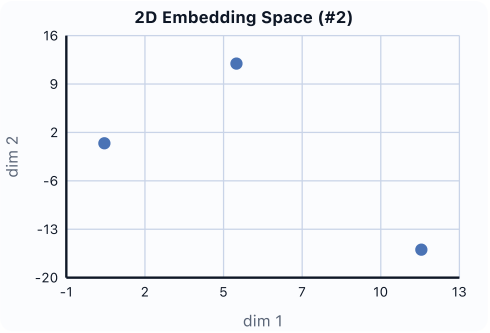
B $\frac{63}{65}$

C $\frac{4}{9}$

D $\frac{63}{130}$

23 MCQ

Compute the cosine similarity $\cos(\theta) = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|_2\|\mathbf{v}\|_2}$ between embedding vectors $\mathbf{u} = [5, 12]^\top$ and $\mathbf{v} = [12, -16]^\top$ (embedding pair #2).



A $-\frac{33}{130}$

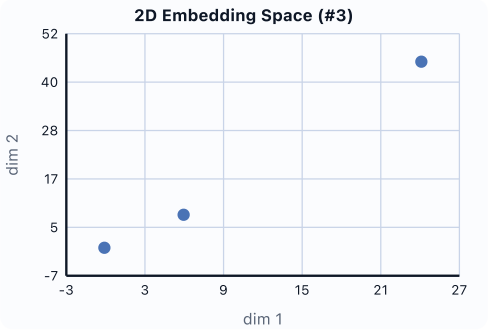
B $-\frac{33}{65}$

C $\frac{11}{23}$

D $\frac{69}{130}$

24 MCQ

Compute the cosine similarity $\cos(\theta) = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|_2\|\mathbf{v}\|_2}$ between embedding vectors $\mathbf{u} = [6, 8]^\top$ and $\mathbf{v} = [24, 45]^\top$ (embedding pair #3).



A $\frac{14}{27}$

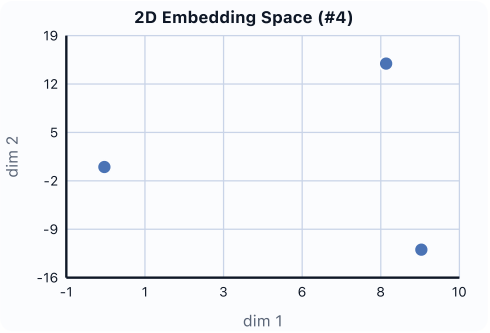
B $\frac{42}{85}$

C $\frac{171}{170}$

D $\frac{84}{85}$

25 MCQ

Compute the cosine similarity $\cos(\theta) = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|_2\|\mathbf{v}\|_2}$ between embedding vectors $\mathbf{u} = [8, 15]^\top$ and $\mathbf{v} = [9, -12]^\top$ (embedding pair #4).



A $\frac{37}{85}$

B $-\frac{36}{85}$

C $-\frac{18}{85}$

D $\frac{17}{32}$